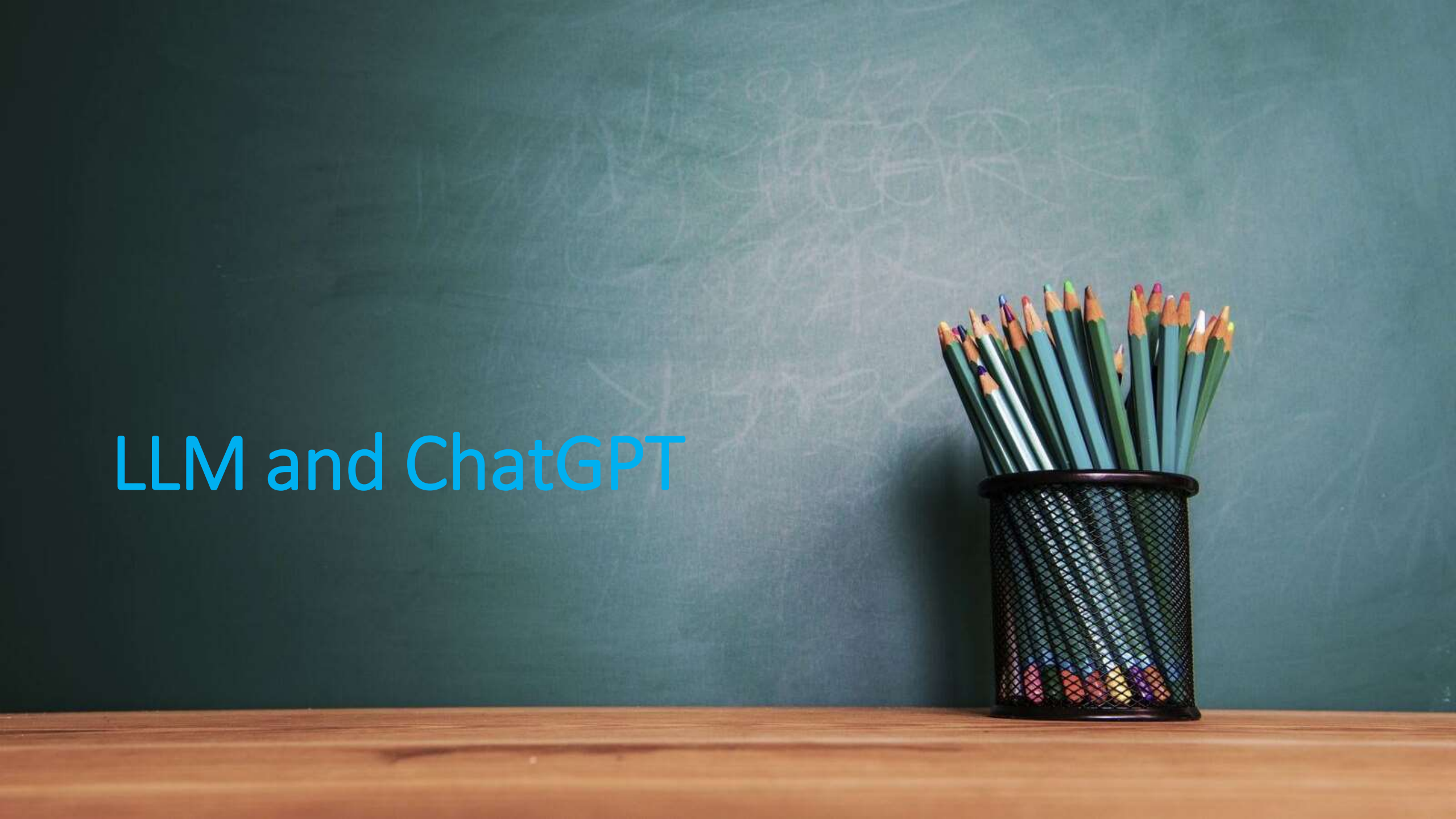
LLM and ChatGPT



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What is Generative AI?

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What is Generative AI

- Generative AI is artificial intelligence capable of generating text, images, or other data using generative models, often in response to prompts.

- Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics

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Transformers and BERT

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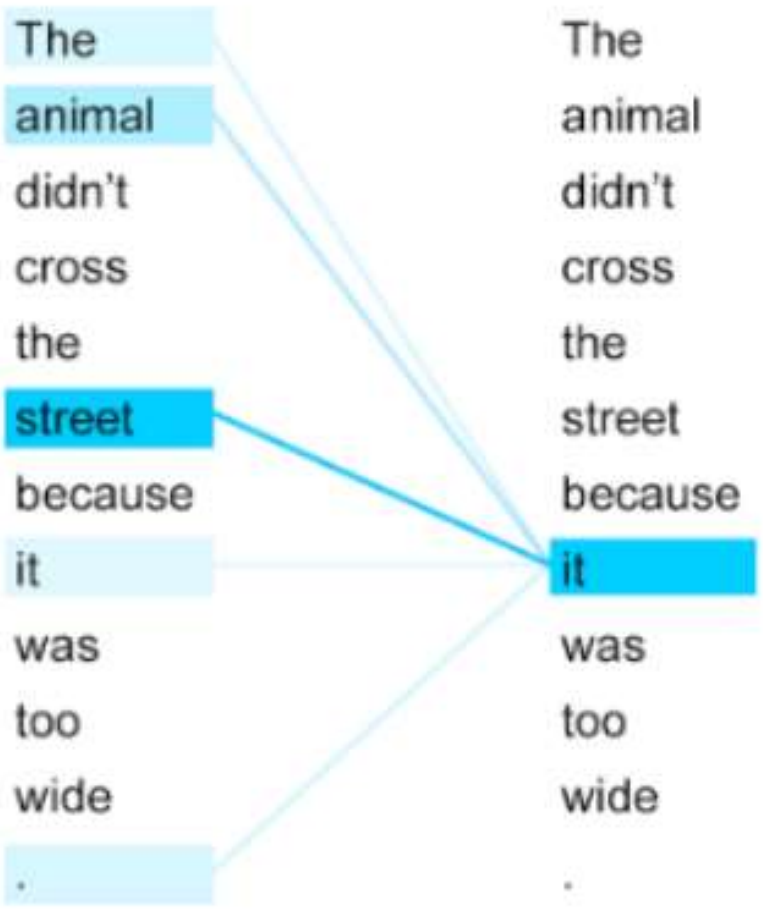
Transformers

- **A transformer is a type of deep learning model and a fundamental building block in many generative AI models**

- It is distinguished by its adoption of self-attention – a Network that focuses on the important parts of the data

- Self-attention captures dependencies between the different parts of a (single) sequence.



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Transformers

Encoder

- Each **encoder layer** helps the model understand which words in the input are important to each other by creating **context-aware representations** of the input..

Decoder

- The **decoder layer** takes the information learned by the encoder and uses it to **generate the output sentence, word by word, using context**

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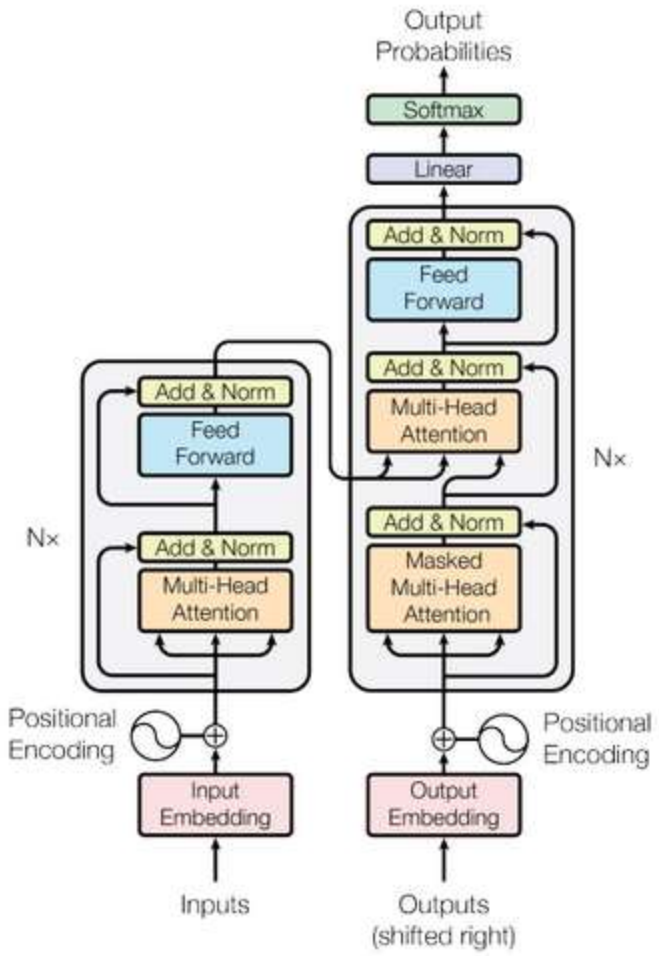

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BERT

- Bidirectional Encoder Representations from Transformers **(BERT) is a language model** introduced in 2018 by researchers at Google
- BERT is based on the transformer architecture.
- BERT was pre-trained simultaneously on two tasks

➢ **Masked word prediction**

Mask some words from the input sentences and predict those words

➢ **Next Sentence prediction**

Given a pair of two sentences along with a label indicating whether they appear one after another in the original corpus model was trained to predict the label

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What Do We Get After Training BERT?

- **BERT learns contextual embeddings:** vector representations of words and sentences that change based on surrounding words.

Feature	Word2Vec	FastText	BERT
Context-aware?	❌ No	❌ No	✅ Yes
OOV Handling	❌ No	✅ Yes (via subwords)	✅ Yes (via WordPiece)
Fixed or Dynamic Vectors?	Fixed	Fixed	Dynamic (contextual)
Word-level or Sentence-level?	Word-level only	Word-level only	Word and Sentence level
Meaning adapts by sentence?	❌ No	❌ No	✅ Yes

What Do We Get After Training BERT?

- **BERT learns contextual embeddings:** vector representations of words and sentences that change based on surrounding words.

Example:
"Bank" in:
- *"He sat by the river bank"*
- *"She opened a bank account"*

Word2Vec / FastText: → Same vector
BERT: → Different vectors (based on context)

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Understanding Language Models

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Overview of Large Language Models (LLMs)

What are Large Language Models?

Large Language Models (LLMs) are advanced **AI systems trained on massive amounts of text data**. They use deep learning, specifically **transformer architectures**, to understand, generate, and manipulate human language.

Examples of Large Language Models (LLMs):

- **GPT series (OpenAI):** GPT-3, GPT-3.5, GPT-4, GPT-5 → Used in ChatGPT.
- **LLaMA (Meta):** Lightweight models for research and open use (LLaMA 2, LLaMA 3).

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Size of language models over time

Release Year	AI Model	Organization	Parameters (Billions)
2025	GPT-5 (projected)	OpenAI	**~2,000 B+ (est.)**
2024	Gemini 1.5 Ultra	Google DeepMind	**~1,500 B (est.)**
2024	Claude 3 Opus	Anthropic	**~1,000 B (est.)**
2023	GPT-4	OpenAI	**~1,000 B (est.)**
2023	LLaMA 2 (70B)	Meta	**70 B**
2023	Mistral Large	Mistral AI	**~45 B (est.)**
2022	GPT-3	OpenAI	**175 B**
2019	GPT-2	OpenAI	**1.5 B**
2018	BERT-Large	Google	**0.34 B**

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What is a prompt?

- A prompt is the input or instruction you give to an AI model (like ChatGPT) to get a desired response.
- It's the question, command, or context you provide to guide how the AI thinks or replies.

- **Example Prompts:**

- "Explain Python functions in simple terms."
- "Write a 100-word email apologizing for a late delivery."
- "Summarize this article in bullet points."

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- **Why It Matters:**

- The **clarity and structure** of your prompt directly affect the **quality of the AI's response**.

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How LLMs process prompts

1. Input Text (Prompt)

- User provides a prompt like: *“W*rite Python code using for loop *“*

2. Tokenization

- Text is broken into **tokens** — subwords or characters. E.g., ["write", " python", " code", " us", "ing", " for", " loop"]

3. Embedding

- Each token is converted into a **numerical vector**. Positional embeddings are added to capture word order.

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How LLMs process prompts

4. Transformer Processing

- The model processes the full sequence using **self-attention**, learning relationships between all tokens in the prompt.

5. Next Token Prediction

- The model predicts the **most likely next token**, one at a time, until the full response is generated.
- Example: After *"The sky is"*, the model predicts "blue" with high probability.

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Even complex outputs are just built by repeating this "next token" prediction step.

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How Can LLMs Write Code or Reason Just by Predicting the Next Token?

- **Massive Training Data**
  LLMs are trained on billions of text and code examples — books, articles, forums, GitHub repos, etc.
- **Pattern Learning**
  They learn not just words, but how ideas are structured — logic, syntax, and reasoning patterns.
- **Chain-of-Thought Reasoning**
  They're trained (or prompted) to think step-by-step, improving reasoning and code generation.

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How Can LLMs Write Code or Reason Just by Predicting the Next Token?

- **Self-Attention Mechanism**
  Every token prediction considers the **entire context**, not just the previous word.

- **Statistical Brilliance**
  They don't understand in the human sense — but they're **very good at guessing** what a correct and useful next token should be.

Result: Accurate answers, working code, and deep explanations — all built **one token at a time**

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How ChatGPT was trained?

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How ChatGPT was trained?

- Generative training – predict next word (GPT 5.2 /GPT 5.2 pro model)
- Supervised fine-tuning – Q and Answers data
- Reward model from human feedback
- Reinforcement learning from human feedback

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Generative training – predict next word

- Input data consists of data from the internet, Wikipedia, Research papers, books etc.
- Model is trained to predict next word /sentence given last n words in the sequence

$P(X_{t+1} = x_{t+1} \mid x_t, x_{t-1}, \ldots\ldots, x_2, x_1)$

Boy was walking along the ___

x_{t+1} (next word in the sentence)	$P(X_{t+1} = x_{t+1})$ (probability)
river	0.4
road	0.2
lake	0.1

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Supervised fine tuning – Q and A data

- To get a model with human like conversational ability, dataset with Question-and-Answer pairs (conversations) generated by human labellers was used

- Model was trained to predict answer for given question (conversation)

- This step helps model to predict conversation like output instead of just predicting next sentence /word

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Reward model

- The model from SFT was tested on the held-out dataset, and multiple prediction outputs were generated from the model for the same question

- These outputs were ranked by human labellers in descending order

- Assume A, B, C, D represent 4 responses from the model

C > A > D> B

- On this labelled data, a reward model is trained to predict the human preferred output (in this case, C)

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Reinforcement learning - PPO

- A new example is sampled from the dataset, and the model (learned policy) generates output for it, and the Reward model calculates the reward for this response
- Calculated reward value is used to update the policy using **Proximal policy optimization (PPO)**
- PPO is a Reinforcement Learning algorithm used to train machine learning models. It is designed to optimize policy functions that map observed states to actions in an environment.
- The main objective of PPO is to iteratively improve the policy of an agent by maximizing the expected cumulative reward obtained from interacting with an environment

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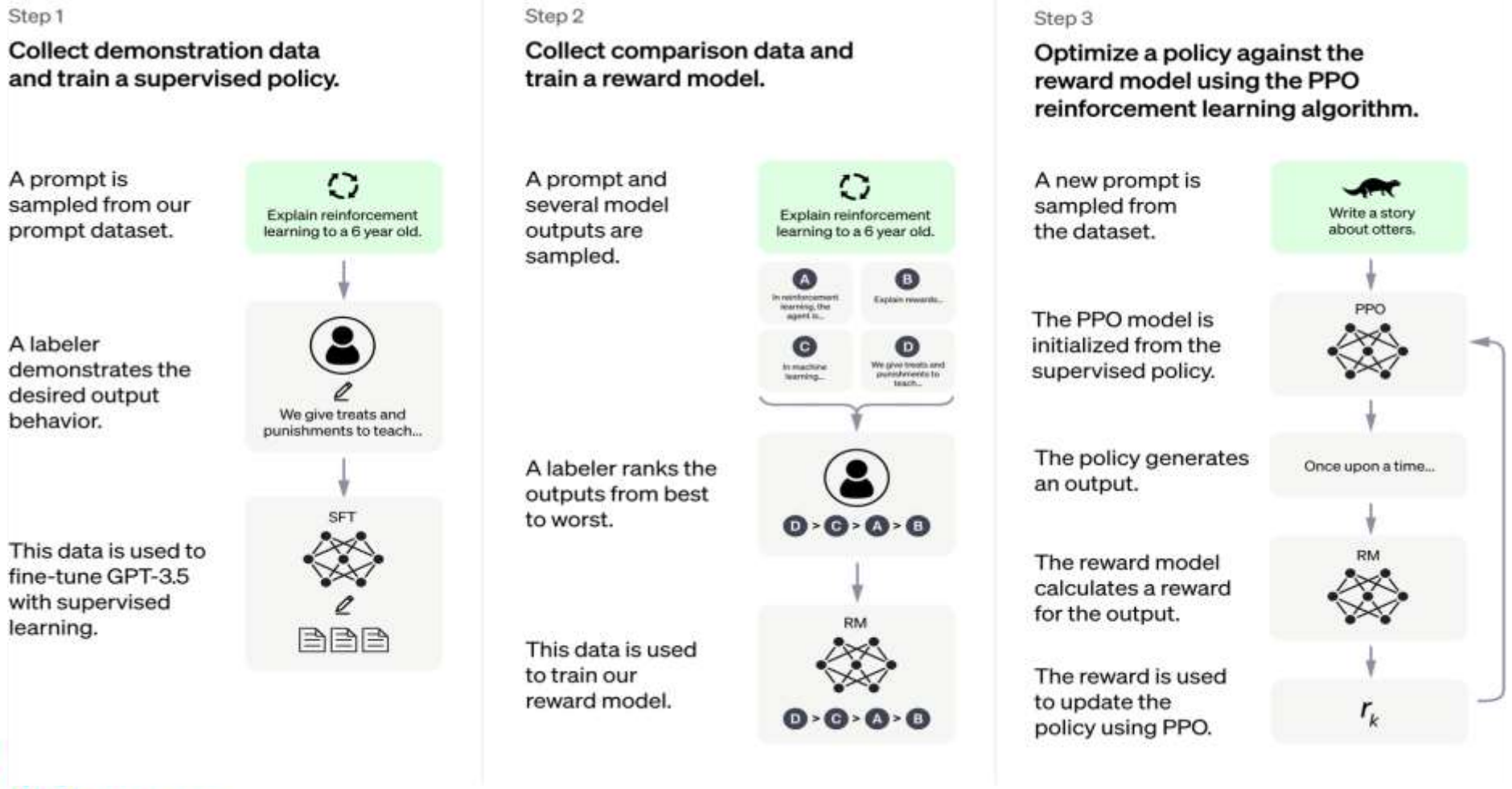
Training of ChatGPT on top of GPT



How to use chatGPT?

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How to use chatGPT

Easiest way to use chatGPT is through web UI provided by openAI

https://chat.openai.com/

You can give instructions / ask questions /chat with the model in the form of prompts

Prompts are a set of instructions we give AI or chatbots, such as ChatGPT, to perform a task.

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Talk with chatGPT

ChatGPT 5.2

How can I become greta Data Scientist. Tell me in bullet points. Done use more than 100 words

- Build strong foundations in **statistics, probability, and linear algebra**
- Master **Python** (Pandas, NumPy, Matplotlib, Scikit-learn)
- Learn **SQL** and basic **data engineering concepts**
- Understand **machine learning algorithms** and when to use them
- Practice **EDA (Exploratory Data Analysis)** deeply
- Work on **real-world projects** and case studies
- Learn to **tell stories with data** (insights > models)
- Understand **business problems**, not just code
- Keep learning: read blogs, papers, and trends
- Communicate clearly with **technical & non-technical teams**

Is this conversation helpful so far?

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Talk with chatGPT

Write a python code for sentiment detection using NLTK

Here's a **simple Python script for sentiment detection using NLTK (VADER)** 👇

python Copy code

```python
# Install required packages (run once)
# pip install nltk

import nltk
from nltk.sentiment import SentimentIntensityAnalyzer

# Download VADER lexicon (run once)
nltk.download('vader_lexicon')

# Initialize sentiment analyzer
sia = SentimentIntensityAnalyzer()

# Sample text
texts = [
    "I love learning Data Science!",
    "This course is okay, not great.",
    "I am very disappointed with the results."
]

# Sentiment analysis
for text in texts:
    score = sia.polarity_scores(text)
    print(f"Text: {text}")
    print(f"Sentiment Score: {score}")

    if score['compound'] >= 0.05:
        print("Sentiment: Positive\n")
    elif score['compound'] <= -0.05:
        print("Sentiment: Negative\n")
    else:
        print("Sentiment: Neutral\n")
```

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GPT 5.2 / GPT 5.2 pro API

GPT 5 model can be accessed through API as well

https://platform.openai.com/docs/guides/chat/introduction

Pricing for API usage is $1 to $21 per 1 M tokens based on the model

Details of pricing can be found at https://openai.com/api/pricing/

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NLP use cases using GPT models

- Sentiment analysis
- Feature extraction from text data
- Chatbot design
- Text summarization
- Text generation on given topic

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Generative AI Models

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Generative AI Models

Model Name	About
ChatGPT	Human like conversational model developed by OpenAI
AutoGPT	Auto-GPT is an "AI agent" that given a goal in natural language, will attempt to achieve it by breaking it into sub-tasks and using the internet and other tools
GitHub CoPilot	Developed by GitHub to assist users of VS code , Jetbrains IDE etc. for autocompleting code
Dall E	DALL-E and DALL-E 2 are deep learning models developed by OpenAI to generate digital images from natural language descriptions
Gemini AI	Multimodal model developed by Google. It can take image, video, audio and text as input

THANK YOU !

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